0. Introduction
1. Nature of individual life: Death process
2. Nature of individual life: Childbearing
3. Characteristics of population that arise from the nature of individual's life
4. Traits of individuals defined in the population
5. Synthesis of the interdisciplinary science

Who gave birth to this baby?

What age parents give birth to the newborns $n_0(t)$?

The breakdown of $n_0(t)$ is

$$n_0(t) = b_1 n_1(t) + b_2 n_2(t) + b_3 n_3(t) + \cdots$$

One baby, chosen arbitrarily, was born from what age parent?

The breakdown of the possibility is

$$1 = \underbrace{\frac{b_1 n_1(t)}{n_0(t)}}_{\text{The degree to which its parent is age-1}} + \underbrace{\frac{b_2 n_2(t)}{n_0(t)}}_{\text{... its parent is age-2}} + \underbrace{\frac{b_3 n_3(t)}{n_0(t)}}_{\text{... its parent is age-3}} + \cdots$$

The simplified ideal population

✓ Change

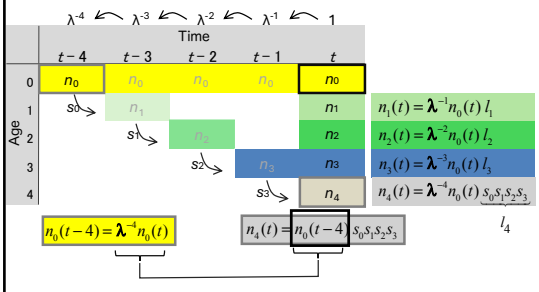
$$n(t) \propto g\lambda^t \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_m \end{pmatrix}$$

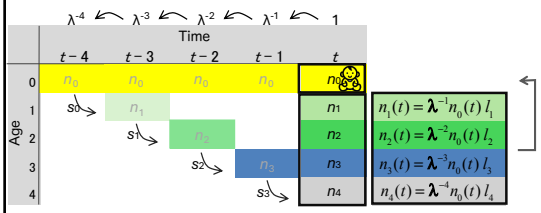
The population changes by a factor of λ in every unit of time.

✓ Compositio n

$$n(t) \propto g\lambda^t \begin{pmatrix} l_0 \\ l_1\lambda^{-1} \\ \vdots \\ u_m\lambda^{-m} \end{pmatrix}$$

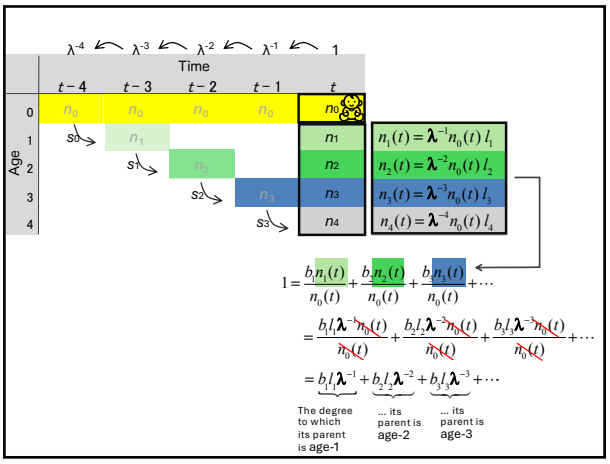
A latent age structure.

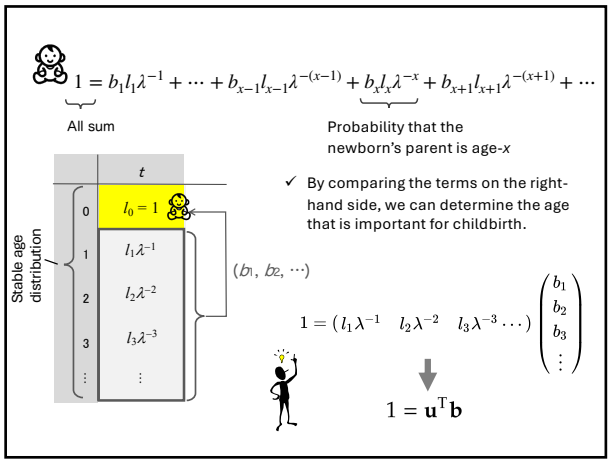


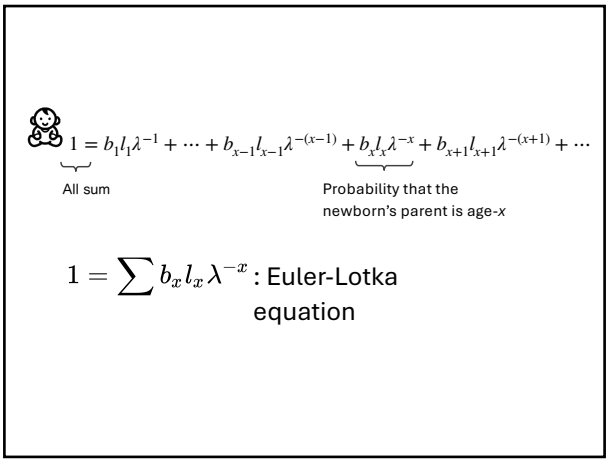


Arbitrary baby was born by a parent at any age.

$$1 = \frac{b_1 n_1(t)}{n_0} + \frac{b_2 n_2(t)}{n_0} + \frac{b_3 n_3(t)}{n_0} + \cdots$$





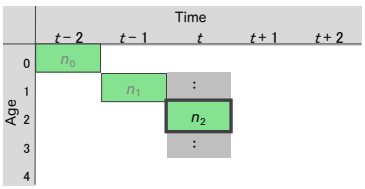


The concept of

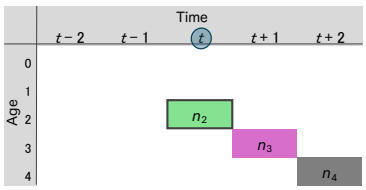
Reproductive value

Relative contribution of an individual of age-*x* for population increase, **in now and in the future**

An age-2 individual at time-*t* contributes to population increase now and, in the future.

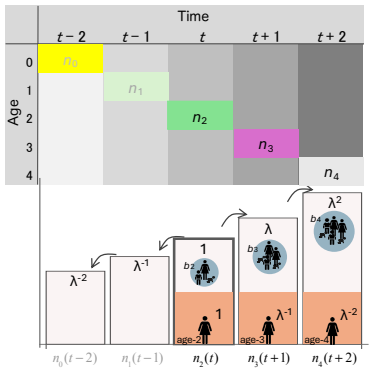


An age-2 individual at time-*t* contributes to population increase now and, in the future.

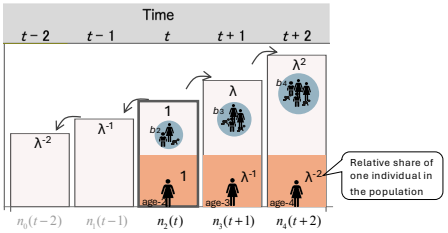


now — In the future

$$b_2 n_2(t) + b_3 n_3(t+1) + b_4 n_4(t+2) + \cdots$$



$$b_2 n_2(t) + b_3 n_3(t+1) + b_4 n_4(t+2) + \cdots$$



Relativized weighting of par capita contribution

$$b_2\lambda^{-0}n_2(t) + \underbrace{b_3\lambda^{-1}}\,n_3(t+1) + b_4\lambda^{-2}n_4(t+1) + \cdots$$

Relative reproductive contribution in one

Sum of the relative reproductive contribution of **n_2 age-2 individuals** now and in the future

Relativized weighting of par capita contribution

$$= b_2\lambda^{-0}n_2(t) + \underbrace{b_3\lambda^{-1}}\,n_3(t+1) + b_4\lambda^{-2}n_4(t+1) + \cdots$$

Relative reproductive contribution in one time future

Sum of the relative reproductive contribution of **one age-2 individual** now and in the future

$$= \frac{b_2\lambda^{-0}n_2(t) + b_3\lambda^{-1}n_3(t+1) + b_4\lambda^{-2}n_4(t+1) + \cdots}{n_2(t)}$$

Sum of the relative reproductive contribution of **one age-2 individual** now and in the future

$$= \frac{b_2\lambda^{-0}n_2(t) + b_3\lambda^{-1}n_3(t+1) + b_4\lambda^{-2}n_4(t+1) + \cdots}{n_2(t)}$$

$$= b_2\lambda^{-0}\frac{n_2(t)}{n_2(t)} + b_3\lambda^{-1}\frac{n_3(t+1)}{n_2(t)} + b_4\lambda^{-2}\frac{n_4(t+1)}{n_2(t)} + \cdots$$

$$= b_2\lambda^{-0}\frac{n_2(t)/n_0(t-2)}{n_2(t)/n_0(t-2)} + b_3\lambda^{-1}\frac{n_3(t+1)/n_0(t-2)}{n_2(t)/n_0(t-2)} + b_4\lambda^{-2}\frac{n_4(t+1)/n_0(t-2)}{n_2(t)/n_0(t-2)} + \cdots$$

$$= b_2\lambda^{-0}\frac{l_2}{l_2} + b_3\lambda^{-1}\frac{l_3}{l_2} + b_4\lambda^{-2}\frac{l_4}{l_2} + \cdots$$

$$= \frac{1}{l_2}\left(b_2\lambda^{-0}l_2 + b_3\lambda^{-1}l_3 + b_4\lambda^{-2}l_4 + \cdots\right)$$

$$= \frac{\lambda^2}{l_2}\left(b_2 l_2\lambda^{-2} + b_3 l_3\lambda^{-3} + b_4 l_4\lambda^{-4} + \cdots\right) = \frac{\lambda^2}{l_2}\sum_{i=2} b_i l_i \lambda^{-i}$$

the future

Reproductive value of age-2 individual $= RV_2 = v_2 = \frac{\lambda^2}{l_2}\sum_{i=2} b_i l_i \lambda^{-i}$

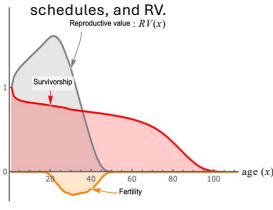
Generally, **reproductive value** of **age-*x*** individual is

$$RV_x = \frac{\lambda^x}{l_x}\sum_{i=x} b_i l_i \lambda^{-i}$$

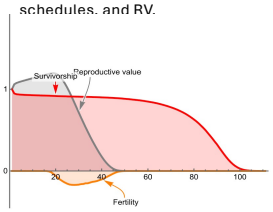
Sum of the relative reproductive contribution of **one age-*x* individual** now and in the future

is called "**r**eproductive **V**alue" of age-*x* $RV_x = \frac{\lambda^x}{l_x}\sum_{i=x} b_i l_i \lambda^{-i}$

Sweden 1891 cohort survival and fertility schedules, and RV.



Sweden 1927 cohort survival and fertility schedules, and RV.



Reproductive value is an important idea to tackle several conceptual issues in Evolutionary Biology.

Comparison of survivorship, fertility, and reproductive value

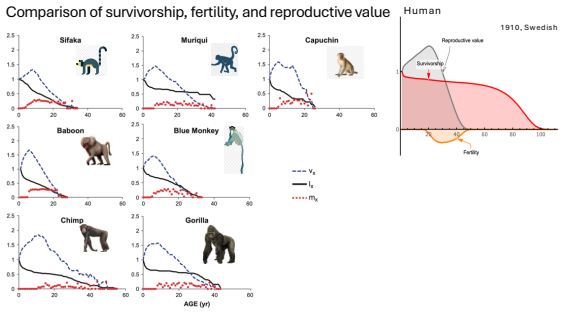


Figure 3. Age-specific survival (l_x), fertility (m_x), and reproductive values (v_x) for seven species of primate. Reproductive value is set to 1 at birth, increases steadily until age of first reproduction, and then declines steadily until the end of life for all species.

Bronikowski 2016



0. Introduction

1. Nature of individual life: Death process
2. Nature of individual life: Childbearing



3. Characteristics of population that arise from the nature of individual's life



4. Traits of individuals defined in the population

Ideal Population's Properties